



Xi'an Jiaotong-Liverpool University

西交利物浦大學

INT305 Machine Learning

Lecture 5

Neural Network and Back Propagation

Sichen Liu

Department Intelligence Science

Sichen.Liu@xjtlu.edu.cn

Neural network

Neural Network: without the brain stuff

(**Before**) Linear score function: $f = Wx$

Neural network

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$

Neural network

第一步: W_1x

第一部分和先前的线性函数一模一样! 这里的 W_1 是第一个权重矩阵。它把输入的 x (3072 个像素值) 转换成一个中间结果。根据图示, 这个中间结果是一个有 100 个数值的向量 h 。我们可以把这 100 个数值看作是网络从原始像素中学到的“特征”或者“模式”。它们不再是原始的像素了, 而是更高层次的组合, 比如“可能存在曲线的边缘”、“可能有一块蓝色的区域”等等。

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

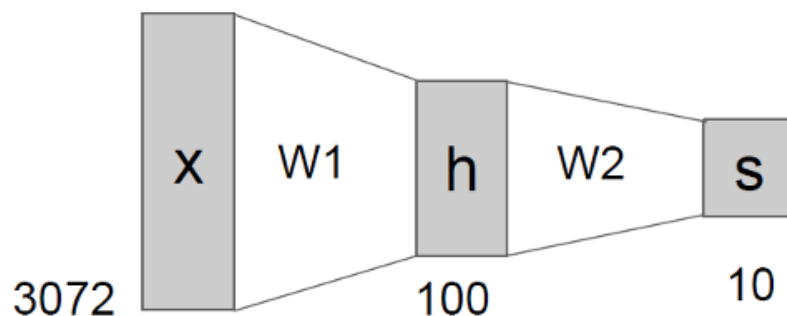
(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1x)$

第三步: W_2

经过 ReLU 激活函数处理后, 我们得到了一个包含了非负特征的向量 h 。

现在, 我们再用第二个权重矩阵 W_2 对这个特征向量 h (100 个值) 进行一次线性变换, 最终得到 10 个类别的分数 s 。

这一步就相当于, 模型根据前面学到的“曲线边缘”、“蓝色区域”等特征, 来做最终的判断: “嗯, 根据这些特征来看, 这张图有 90% 的可能是猫, 10% 的可能是狗。”



如果把 $\max(0, \dots)$ 去掉, 两层神经网络会退化成线性分类器。

第一层($W_1 + \text{ReLU}$): 从原始像素中提取有用的抽象的特征。

第二层(W_2): 根据这些提取出来的特征, 进行最终类打分

第二步: $\max(0, \dots)$

这是它与线性函数最根本的区别! 这个函数叫做激活函数 (Activation Function), 而 $\max(0, z)$ 是其中最常用的一种, 它的名字叫 ReLU (Rectified Linear Unit), 中文叫“修正线性单元”。

它的工作原理超级简单: 如果一个数是正数, 就保持不变; 如果一个数是负数, 就把它变成 0。这是为了引入非线性 (non-linearity)。这就好比我们给了模型一个“开关”, 让它能够学会更复杂的、弯弯曲曲的决策边界。没有这个激活函数, 无论我们叠加多少层线性函数, 最终结果都等价于一个线性函数, 就像把几个直尺首尾相连, 得到的还是一根更长的直尺。但有了 ReLU 这个“关节”, 我们就可以把直尺折叠, 去拟合更复杂的形状!

Neural network

最内层： $\max(0, W_1 x)$

和之前一样，这是我们的第一个隐藏层。它接收原始的图片像素 x ，通过 W_1 计算，再经过 ReLU 激活，输出了第一组比较初级的特征（比如一些边缘、颜色块等）。

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$

or 3-layer Neural Network:

$$f = W_3 \max(0, W_2 \max(0, W_1 x))$$

中间层： $\max(0, W_2 \dots)$

现在是关键的一步！它不再直接把这些初级特征用于最终分类，而是把它们作为输入，再进行一次相同的操作：乘以第二个权重矩阵 W_2 ，然后再次通过 ReLU 激活函数。

这一层就是第二个隐藏层。它的作用是把第一层学到的简单特征（边缘、色块）进行组合，从而学习到更复杂、更有意义的特征（比如“眼睛的轮廓”、“鸟的喙”）。

最外层： $W_3(\dots)$

这是我们的输出层。它接收第二个隐藏层输出的那些复杂的组合特征，然后乘以 W_3 ，最终计算出每个类别的最终分数 f 。

每增加一个“隐藏层”，网络就获得了从上一层特征中学习更高级、更抽象特征的能力。这就是“深度学习”中“深度 (Deep)”的来源——我们通过堆叠更多的层来构建一个更深的网络

★ Activation functions

可以把激活函数想象成每个神经元的“性格开关”或者“决策模型”。它决定了接收到信号后，这个神经元应该被“激活”到什么程度，以及应该向下一层传递什么样的信息。它的核心使命，我们再强调一遍，就是引入非线性，让网络能够学习复杂的数据模式。

Activation Functions

Sigmoid

$$\sigma(x) = 1/(1 + e^{-x})$$

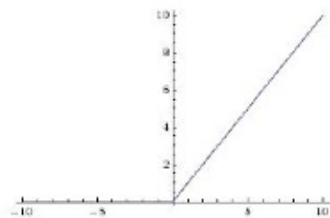
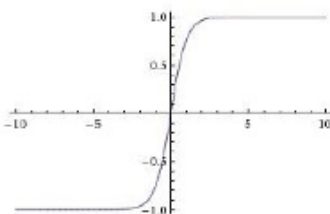
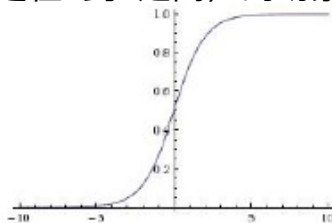
tanh $\tanh(x)$

ReLU $\max(0, x)$

计算超级快，而且在正数区域不会有梯度消失的问题，所以学习得很快。

有个叫“Dying ReLU”的问题。如果一个神经元的输入总是负数，它就会一直输出 0，梯度也永远是 0，这个神经元就再也无法学习了

它能任何输入的数值都“压缩”到 0 和 1 之间。这非常有用，比如在需要输出一个概率值（概率总是在 0 到 1 之间）的场景下。

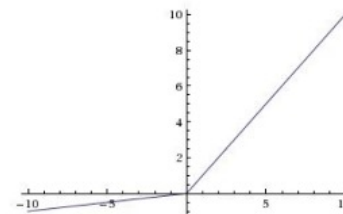


leaky ReLU

当输入是负数时，它不再输出 0，而是输出一个很小的值（比如 $0.1x$ ）。这就好比给“死掉”的神经元做了一点心肺复苏，让它还有机会被重新激活。

Leaky ReLU

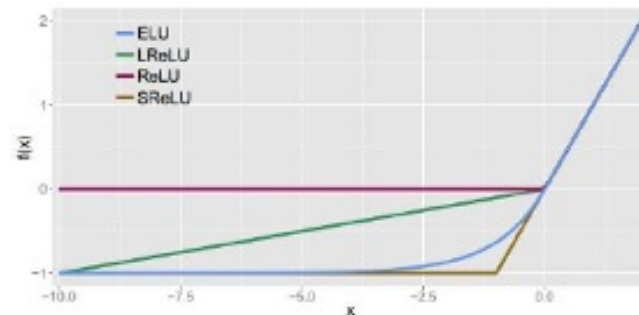
$$\max(0.1x, x)$$



Maxout $\max(w_1^T x + b_1, w_2^T x + b_2)$

ELU

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(\exp(x) - 1) & \text{if } x \leq 0 \end{cases}$$

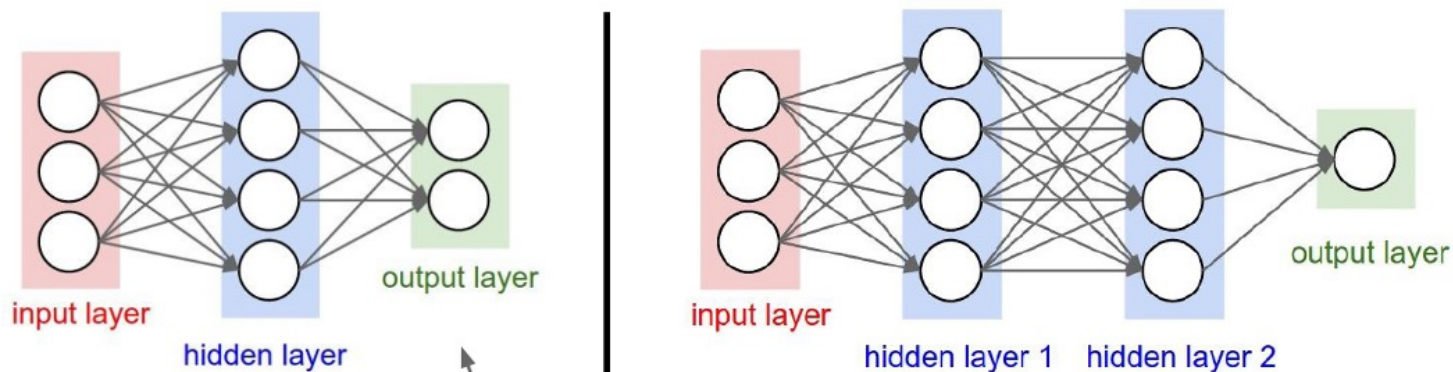


Maxout 这是一个更广义的激活函数，它本身是可以学习的。可以看作是 ReLU 和 Leaky ReLU 的泛化版本，效果很好，但计算开销和参数数量也更大了。

ELU 是 Leaky ReLU 的一个更平滑的升级版，既解决了“Dying ReLU”问题，又具有接近 0 均值的输出。但计算会比 ReLU 复杂一点。

Neural network

Neural Networks: Architectures



"2-layer Neural Net", or
"1-hidden-layer Neural Net"

"3-layer Neural Net", or
"2-hidden-layer Neural Net"

"Fully-connected" layers

一个隐藏层，一个输出层，

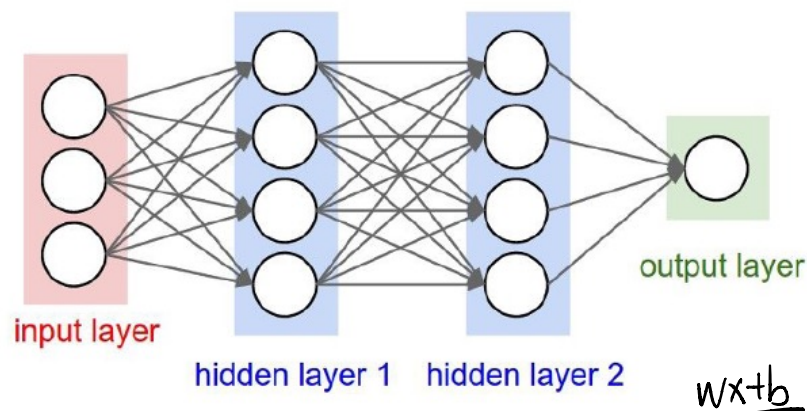
每层之间的节点，必须是完全连接。

前一层的所有神经元，都与后一层所有神经元相连。

两个隐藏层，一个输出层

Neural network

Example Feed-forward computation of a Neural Network



```
# forward-pass of a 3-layer neural network:  
f = lambda x: 1.0/(1.0 + np.exp(-x)) # activation function (use sigmoid)  
x = np.random.randn(3, 1) # random input vector of three numbers (3x1)  
h1 = f(np.dot(W1, x) + b1) # calculate first hidden layer activations (4x1)  
h2 = f(np.dot(W2, h1) + b2) # calculate second hidden layer activations (4x1)  
out = np.dot(W3, h2) + b3 # output neuron (1x1)
```

输入数据

第二个隐藏层

sigmoid 激活函数

→ 第一个隐藏层

输出层



Gradient Descent 梯度下降

Where we are...

$$s = f(x; W) = Wx$$

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$$L = \frac{1}{N} \sum_{i=1}^N L_i + \sum_k W_k^2$$

want $\nabla_W L$

梯度指向了让损失上升最快的方向,那我们要考虑怎么更新权重 w 来降低损失,让模型变得更好.

所以梯度下降就是沿着梯度的反方向走

scores function 对于一个给定的输入,模型对每个可能类别有多大的信心

SVM loss (hinge loss) 用来衡量模型在单个样本上预测得有多差,值越高,说明模型在这一次预测上犯的错误越大

data loss + regularization

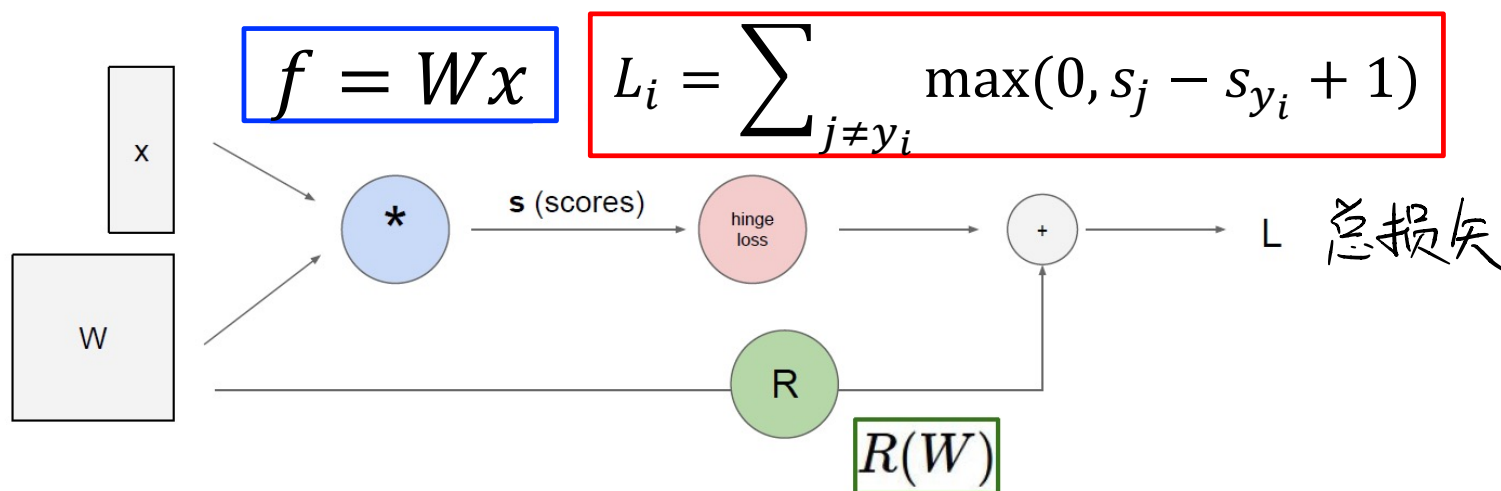
对模型在整个数据集上表现的最终评价

$\sum_{i=1}^N L_i$: Data loss (数据损失): 把所有的单个样本损失 (L_i) 加起来,再求一个平均值。代表了我的模型在整个训练集上的平均不满意度。

$\sum W_k^2$: 惩罚过于复杂的模型。把所有权重 w 的平方加起来,如果权重值太大,这一项就会很大,导致总损失变大。主要用于防止过拟合。

Computational Graph

Computational Graph



Computational Graph

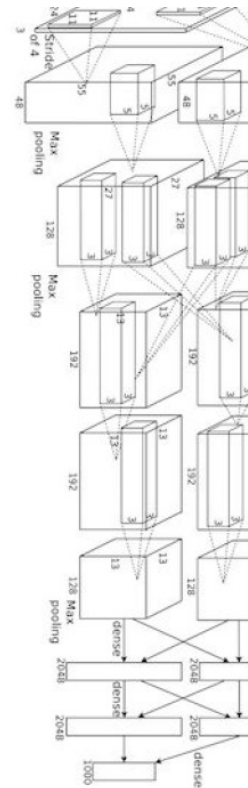
Convolutional Network
(AlexNet)

卷积神经网络

input image

weights

loss

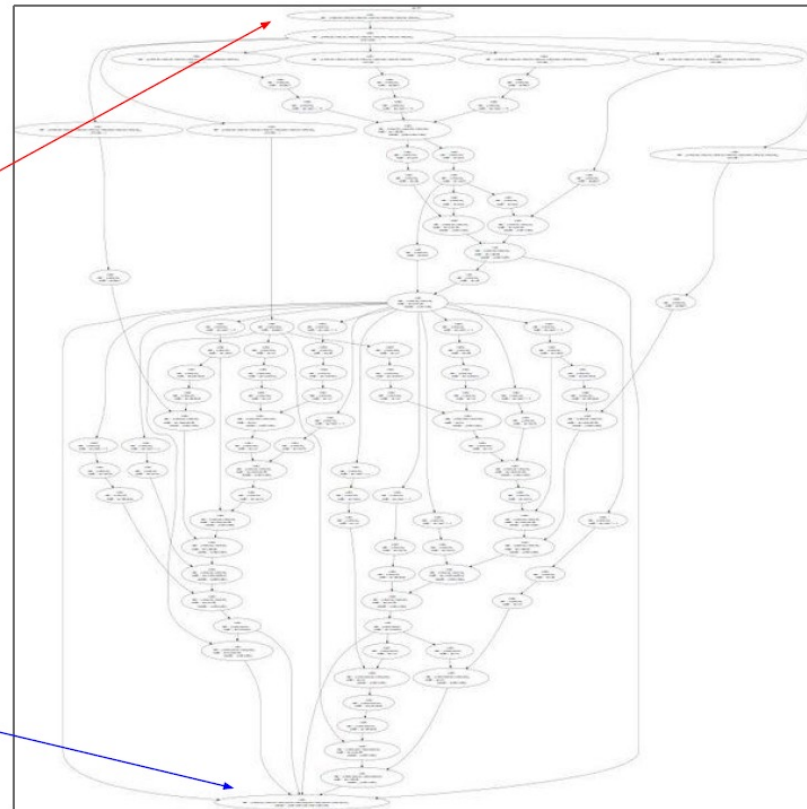


Computational Graph

Neural Turing Machine
神经网络图灵机

input tape

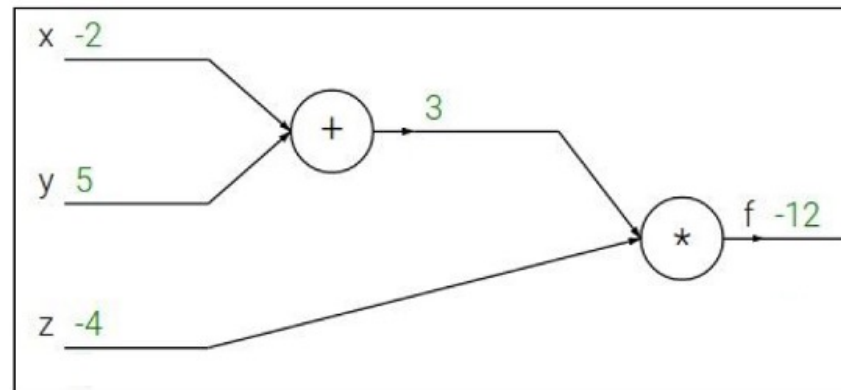
loss



Example 1

$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$



Example 1

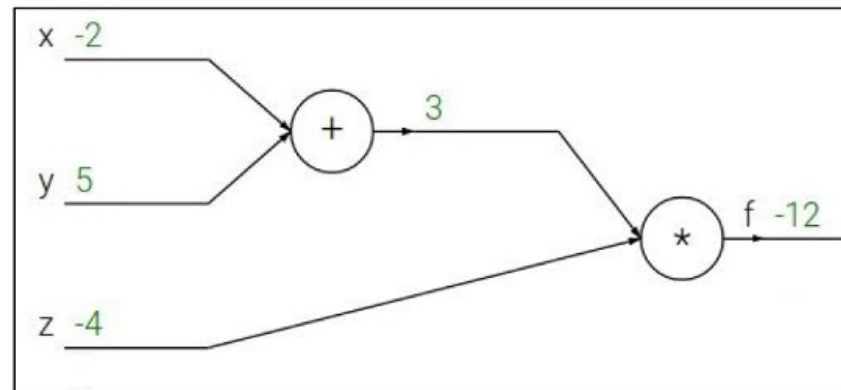
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

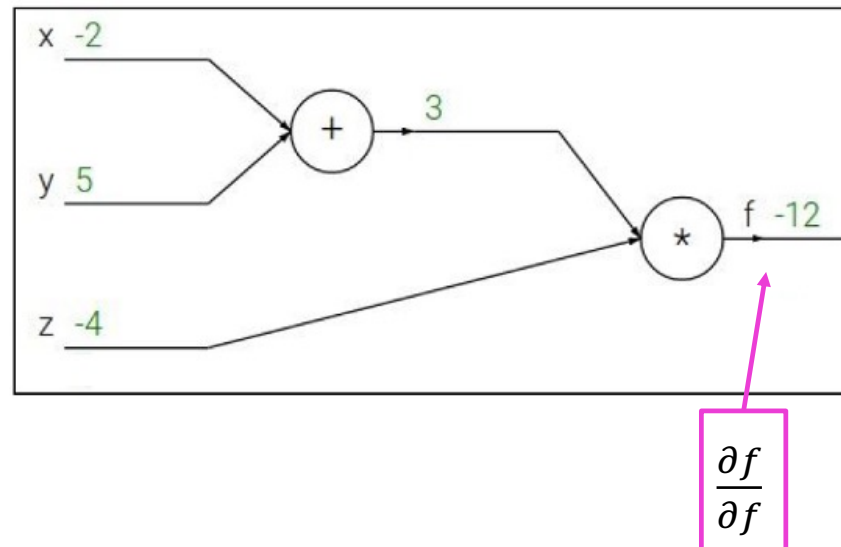
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

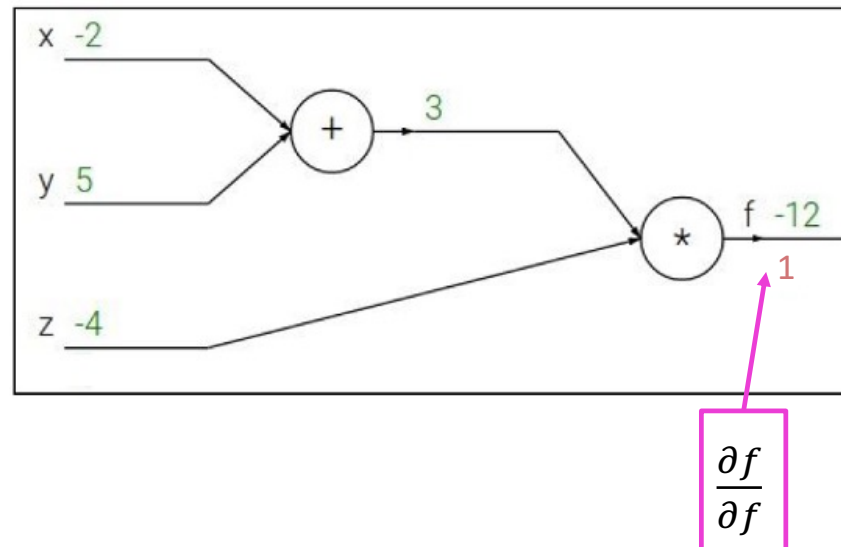
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

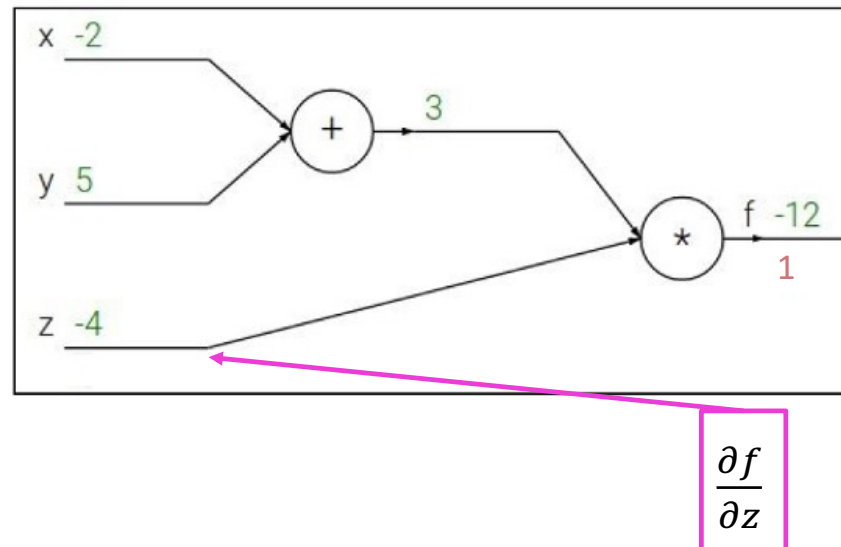
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

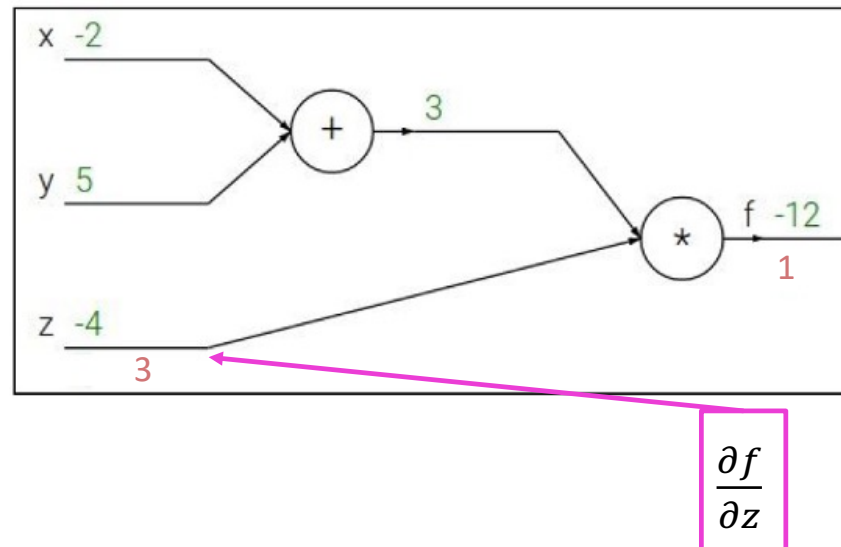
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

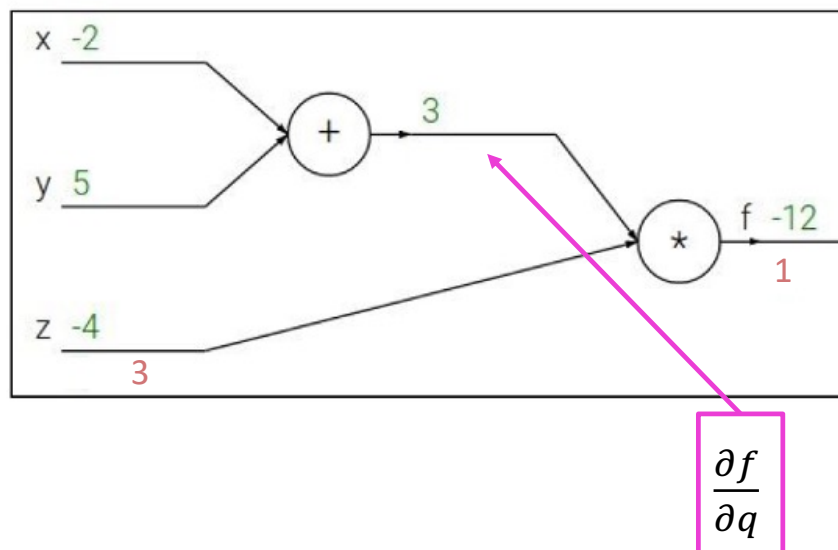
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

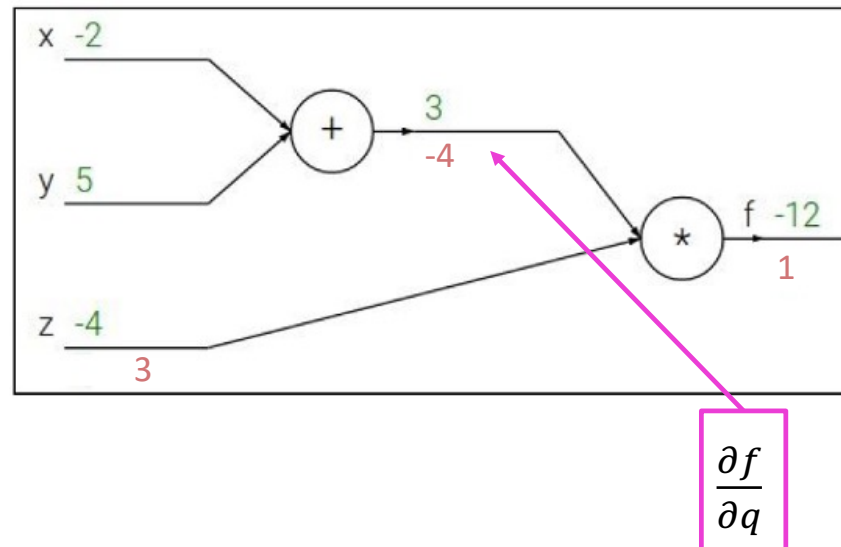
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

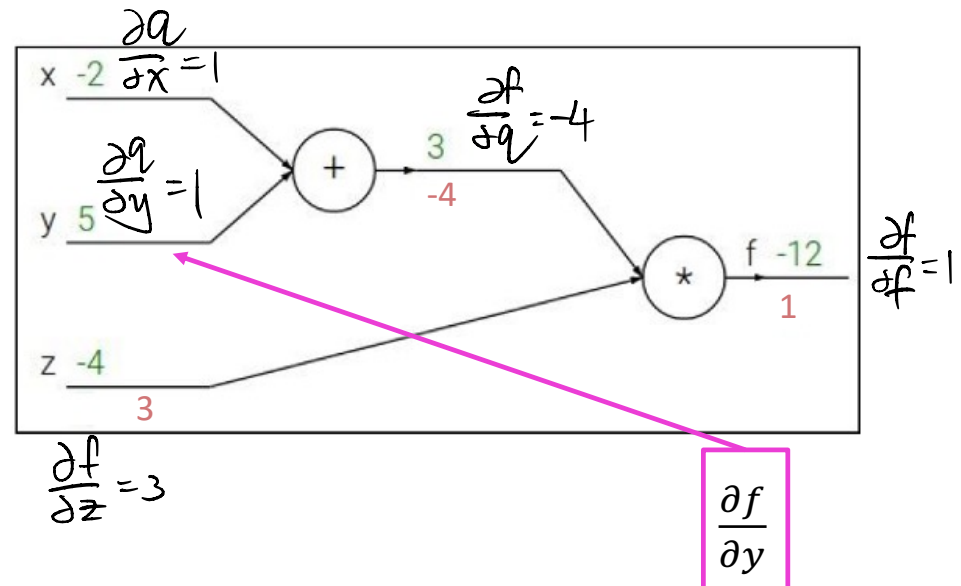
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

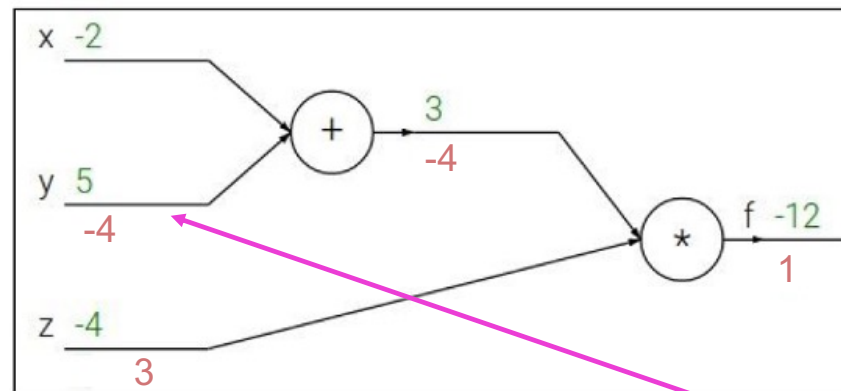
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



$$\frac{\partial f}{\partial y}$$

Chain rule:

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y}$$

$$(-4) \cdot 1 = -4$$

Example 1

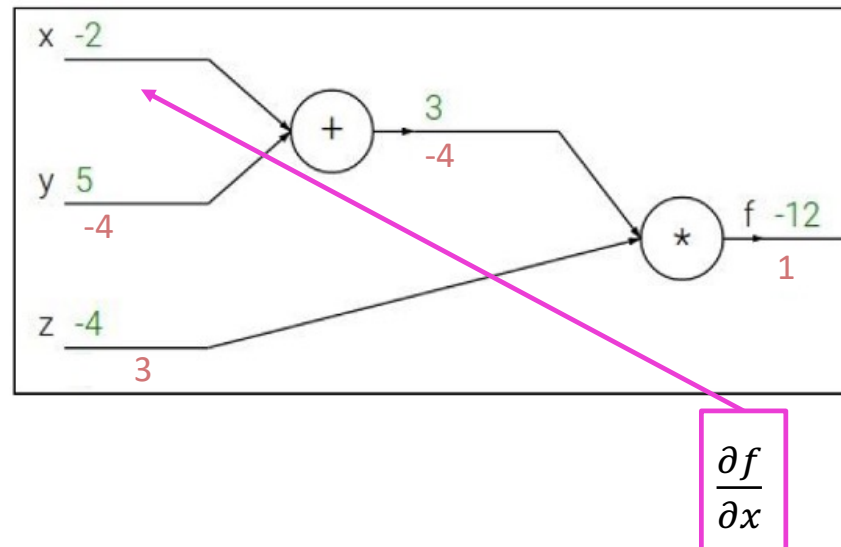
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

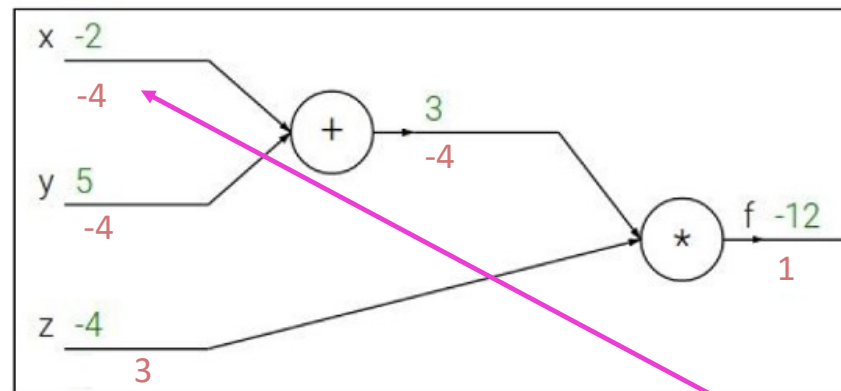
$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$

$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



$$\frac{\partial f}{\partial x}$$

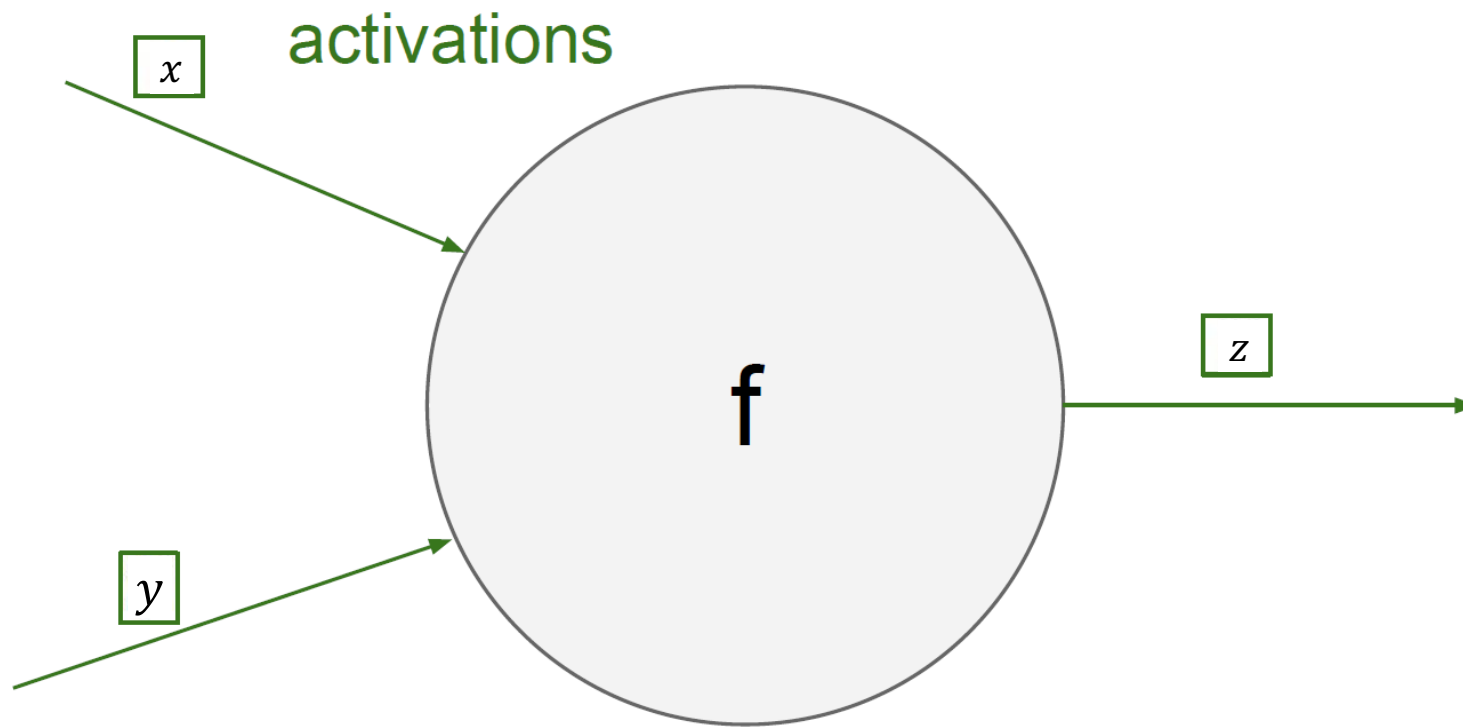
Chain rule:

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y}$$

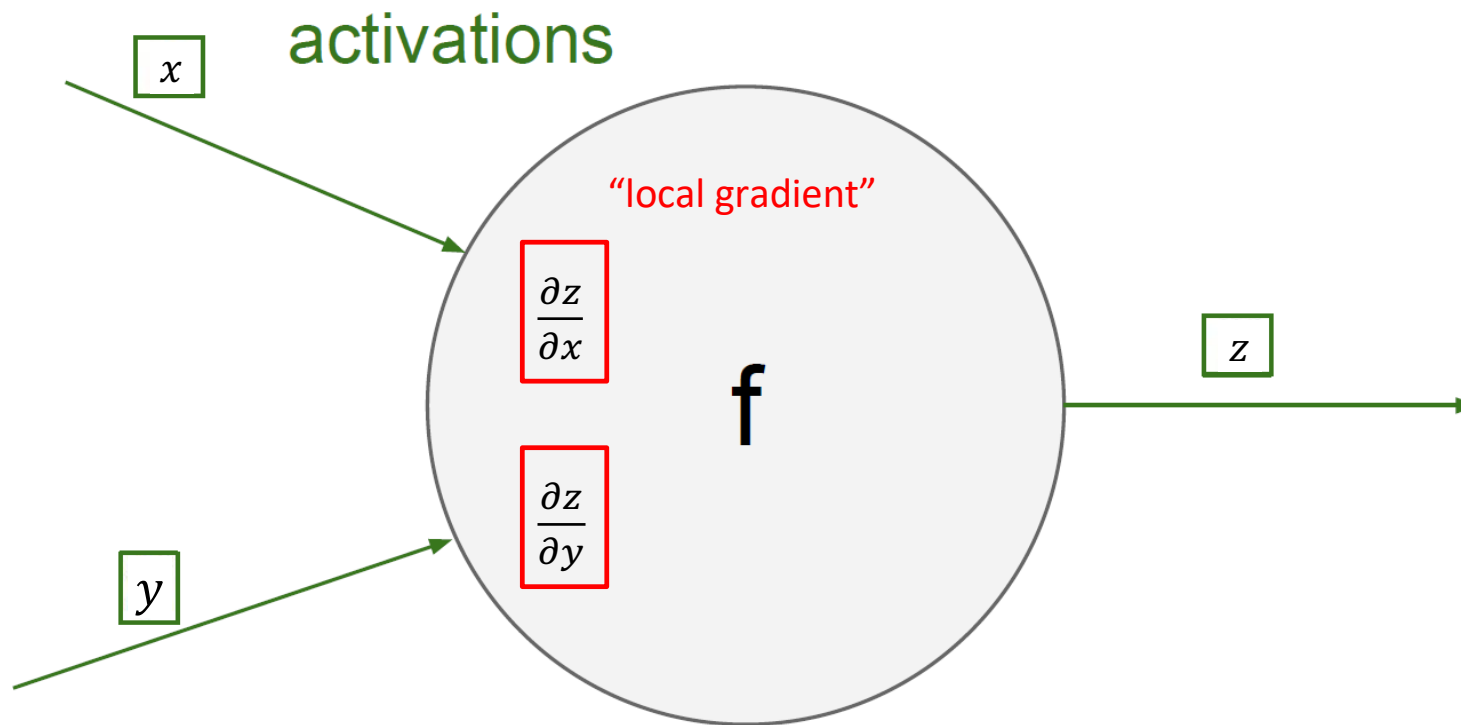
为什么是y?

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x}$$

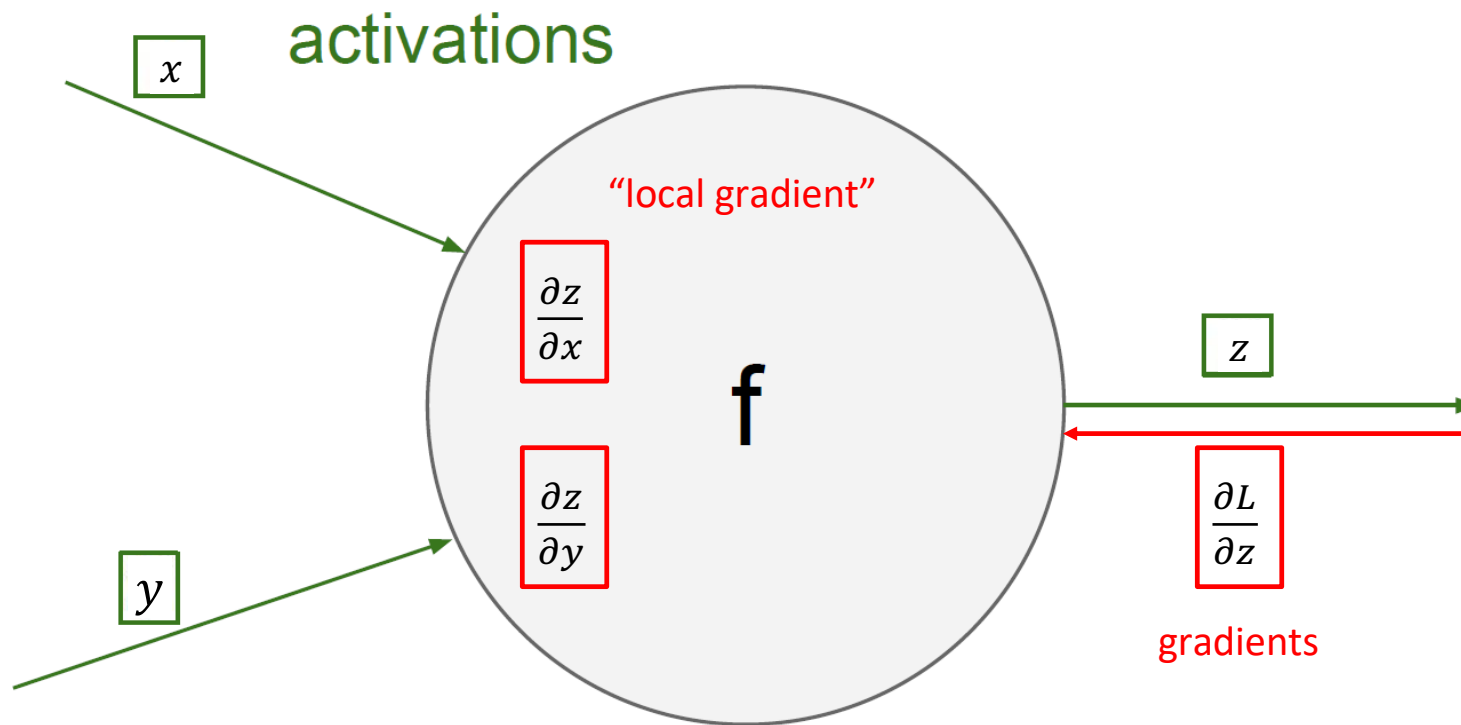
Chain rule



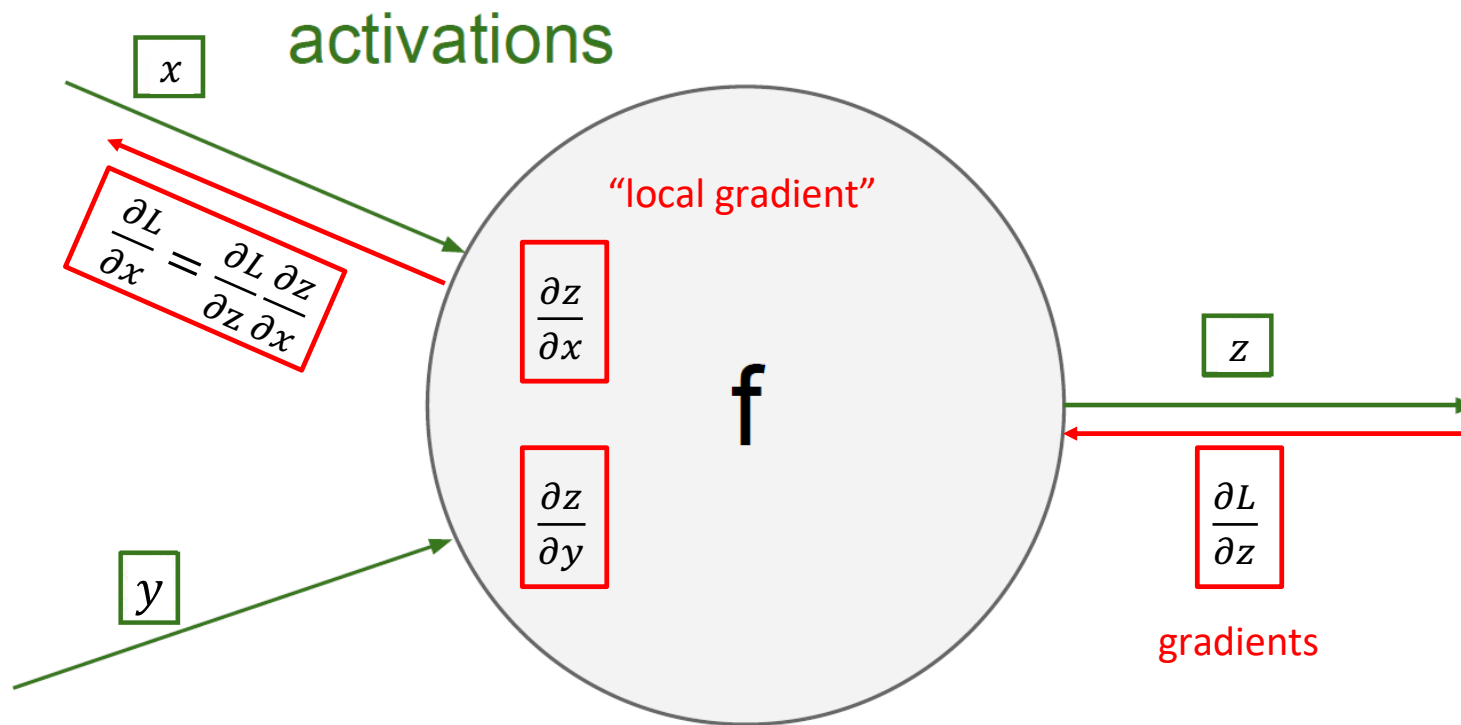
Chain rule



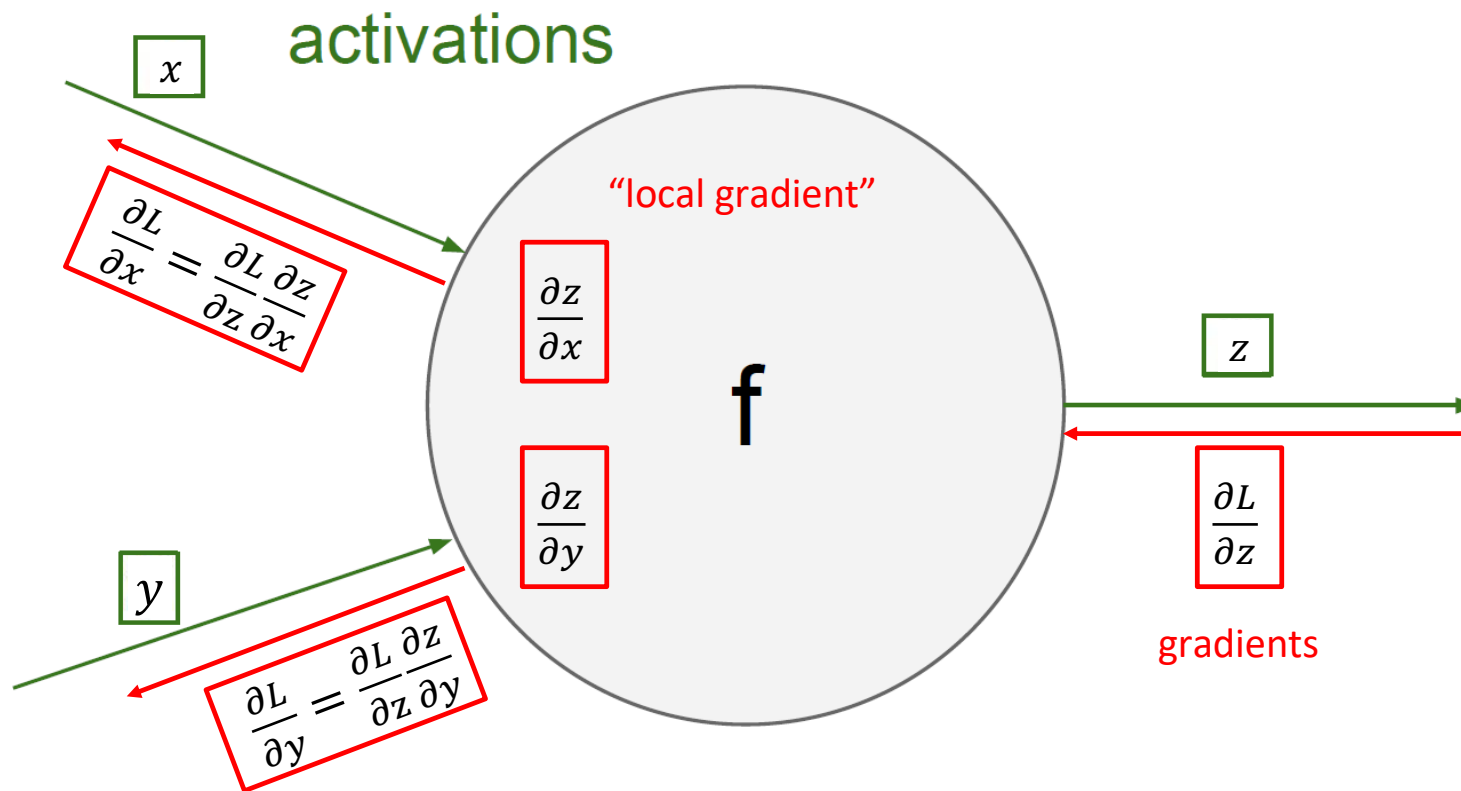
Chain rule



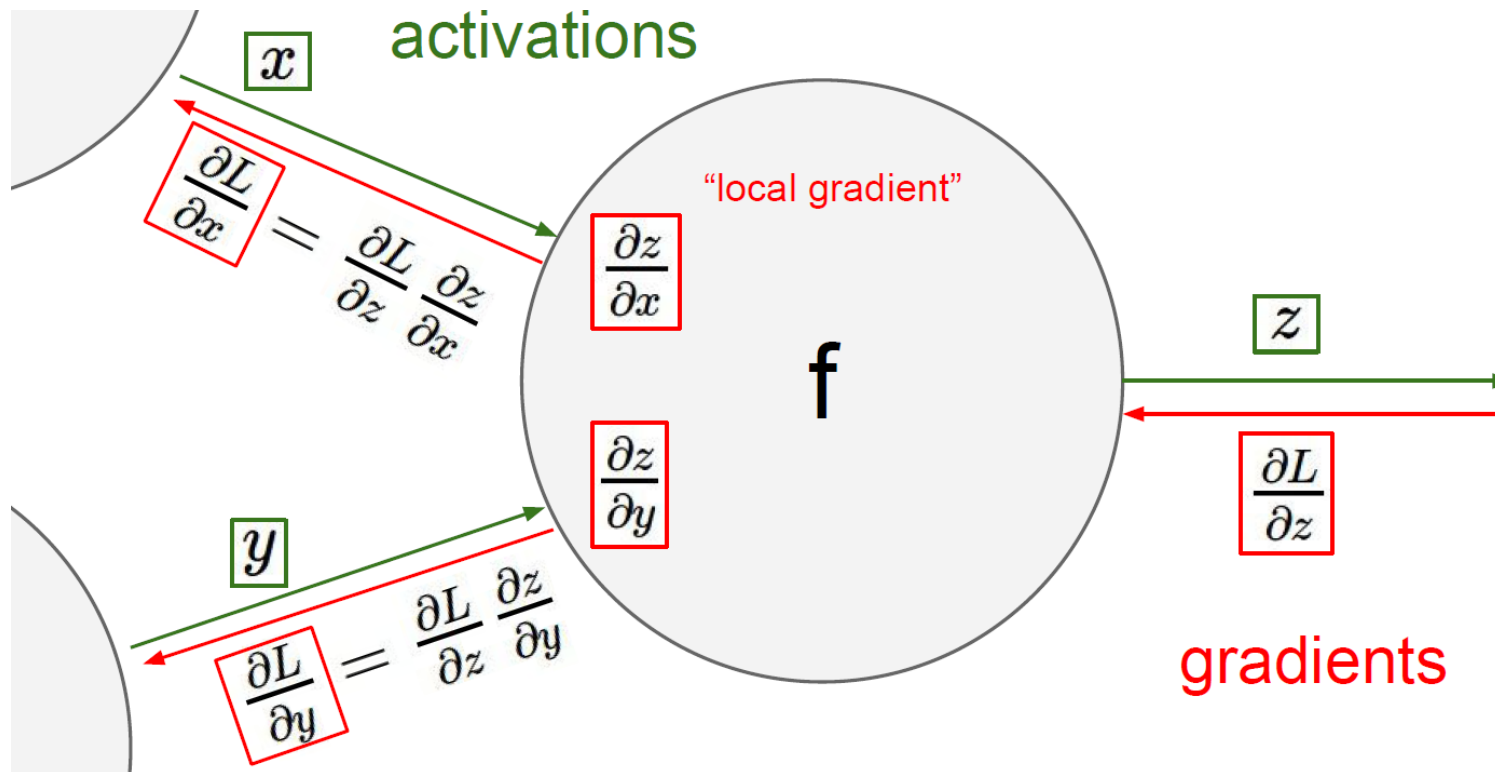
Chain rule



Chain rule

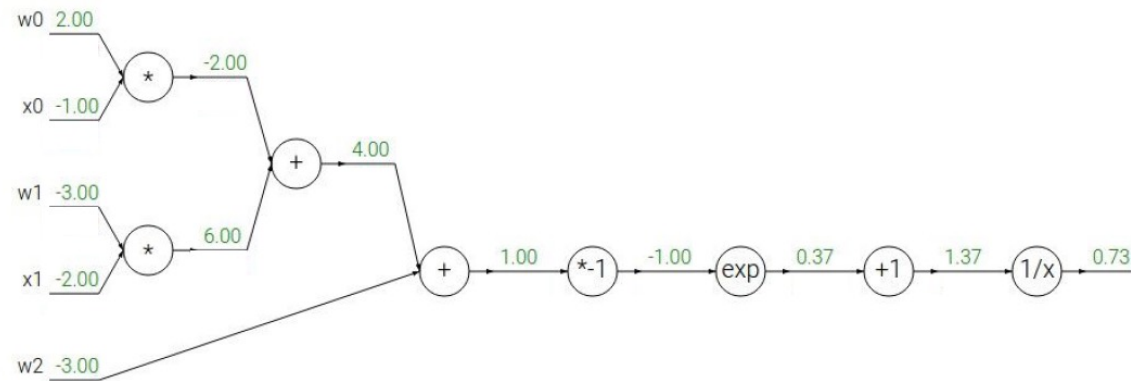


Chain rule



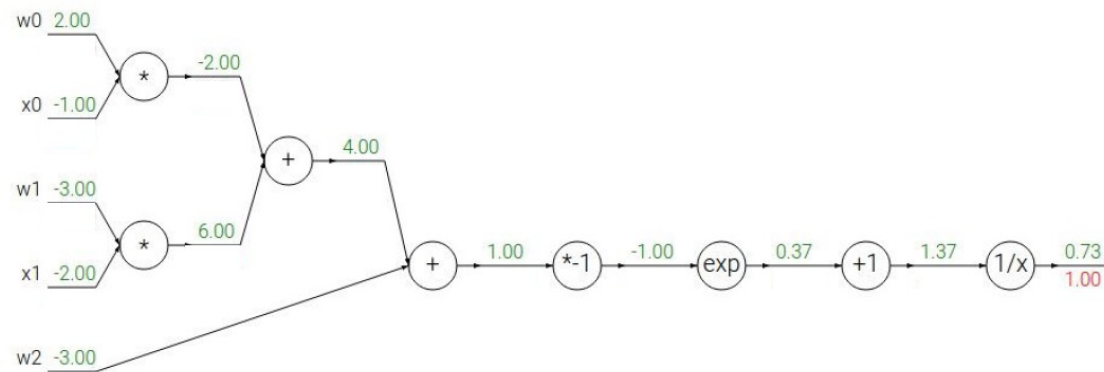
Chain rule

Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$



Chain rule

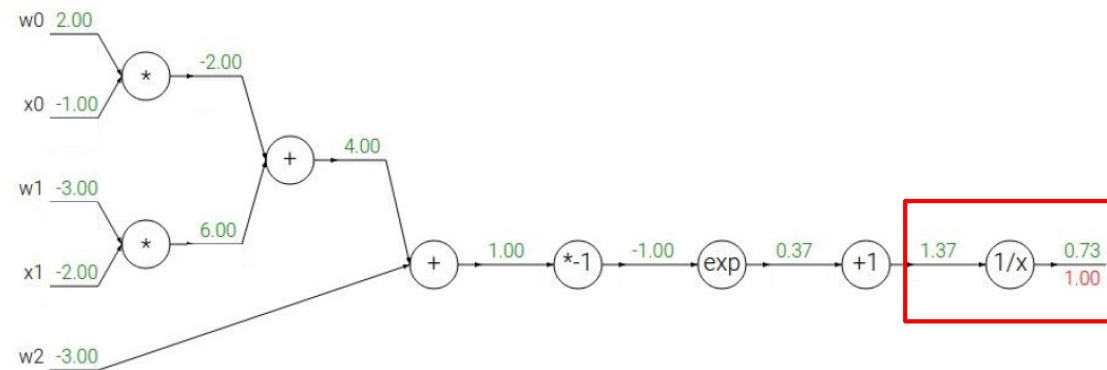
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

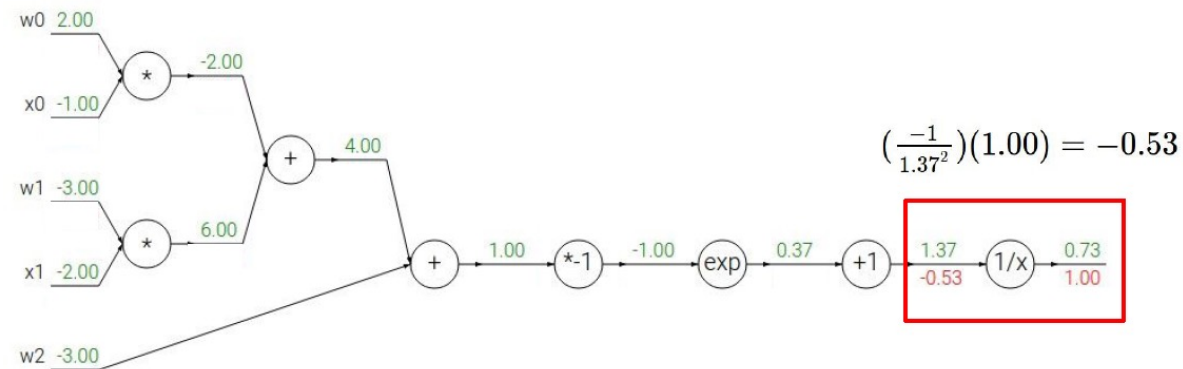
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

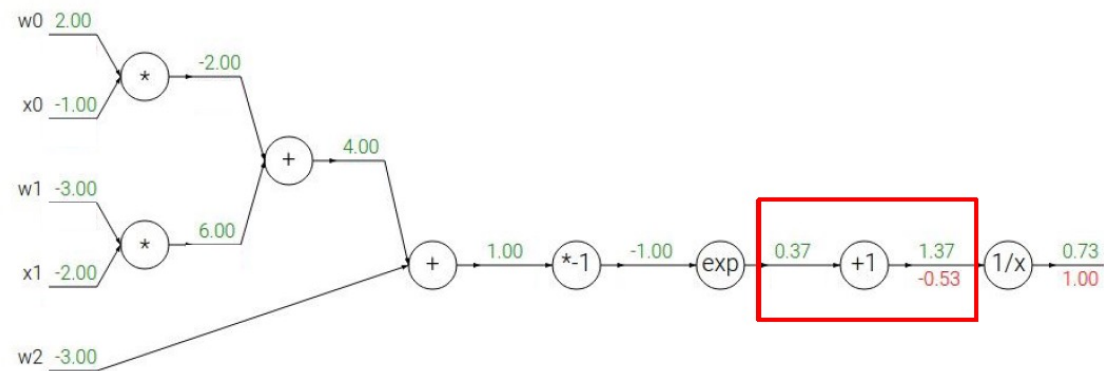
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

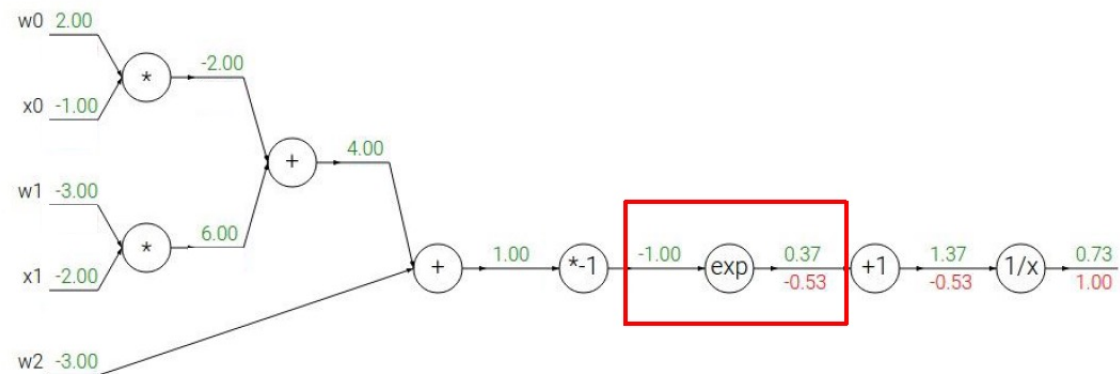
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$



$f(x) = e^x$	→	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	→	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	→	$\frac{df}{dx} = a$		$f_c(x) = c + x$	→	$\frac{df}{dx} = 1$

Chain rule

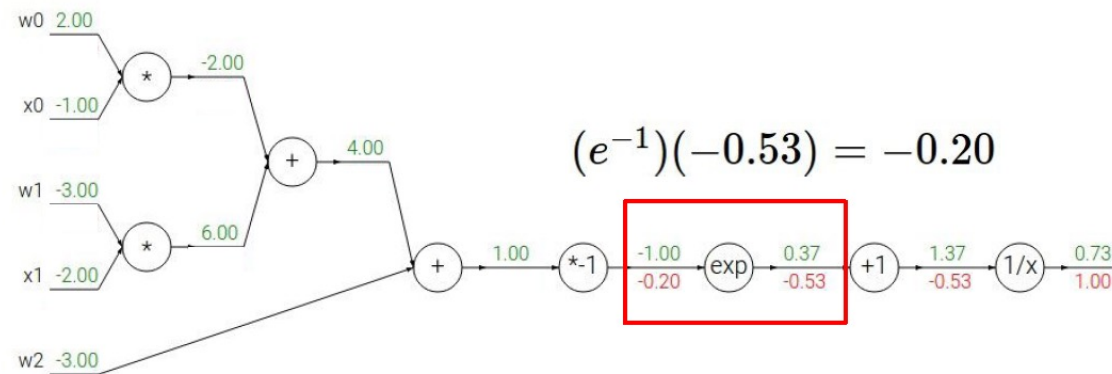
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$



$$(e^{-1})(-0.53) = -0.20$$

$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

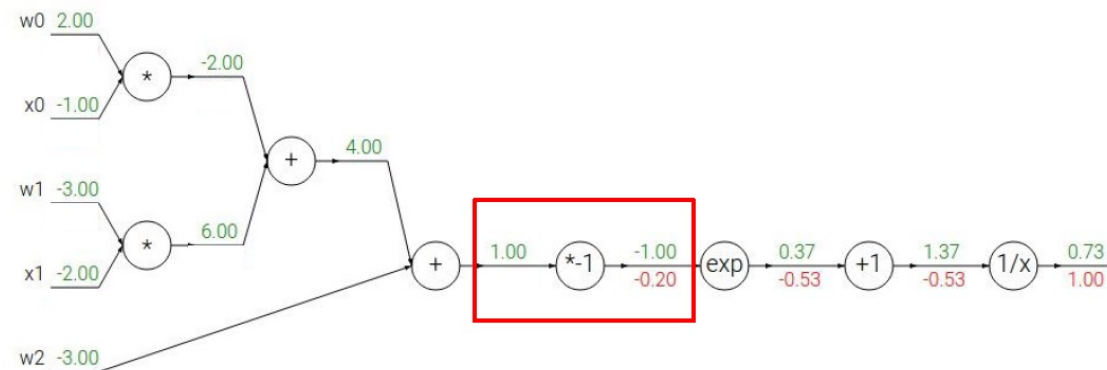
$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

$$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$$

Chain rule

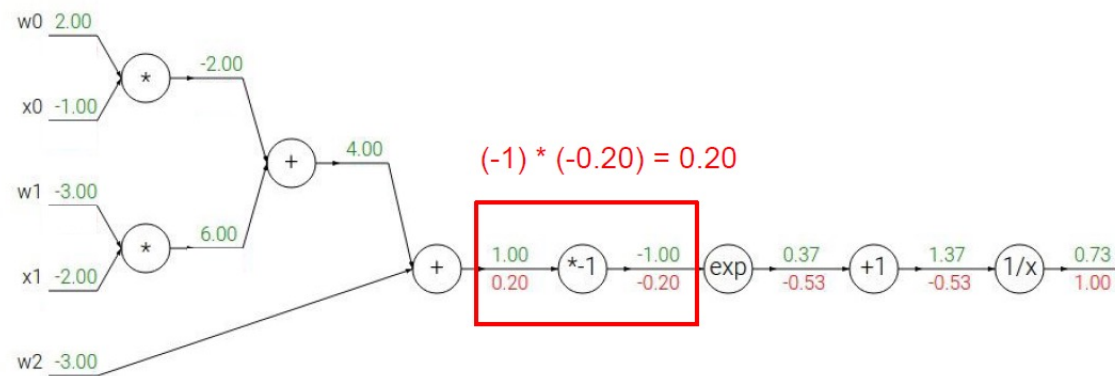
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

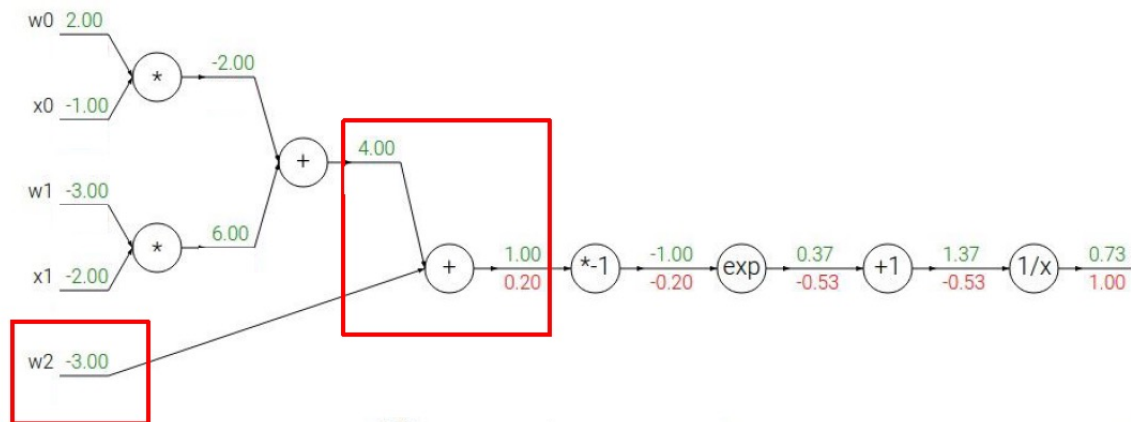
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

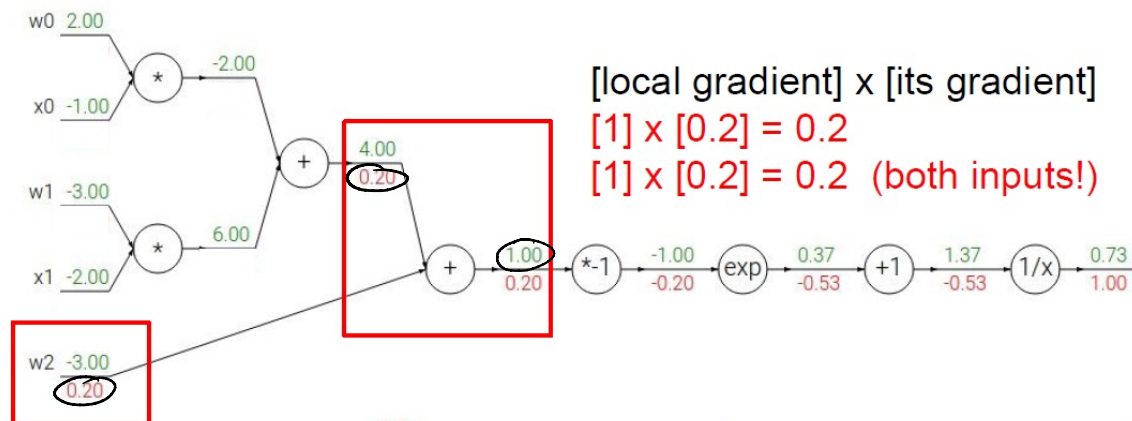
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$

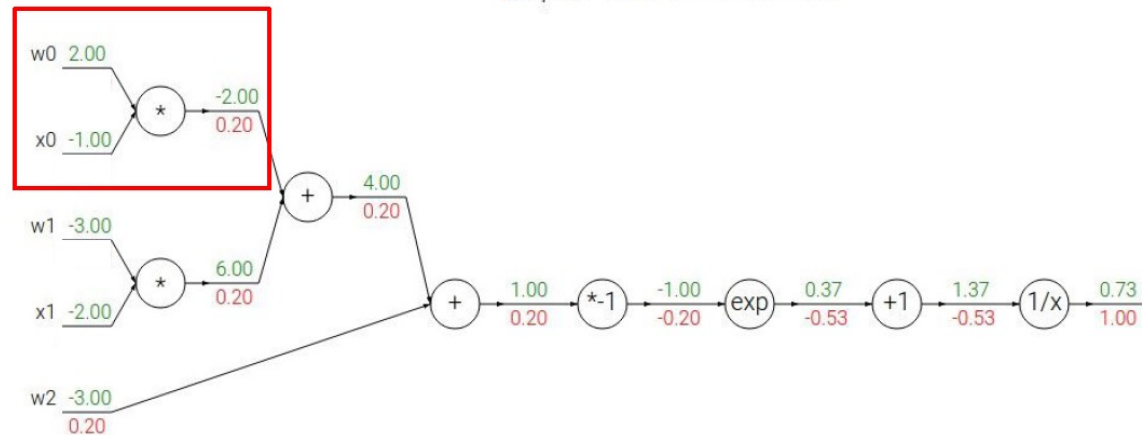


$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

Another example:

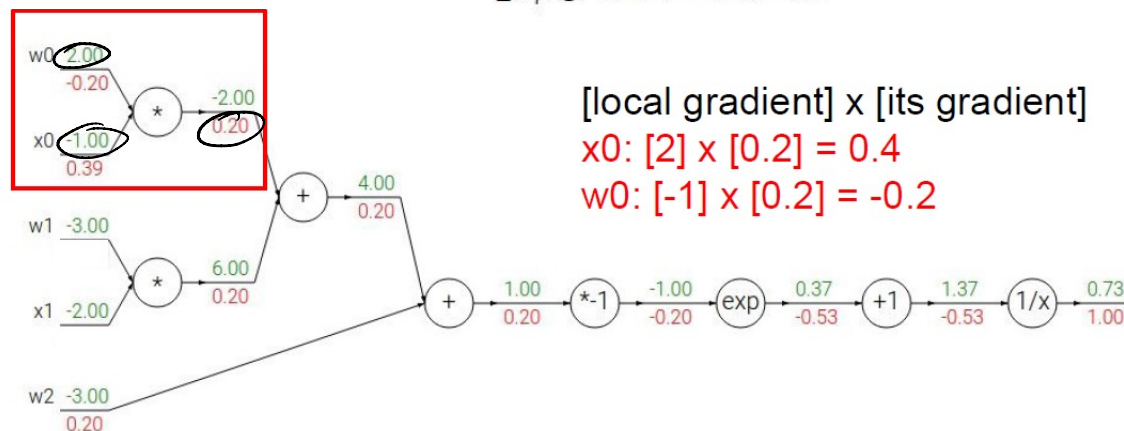
$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



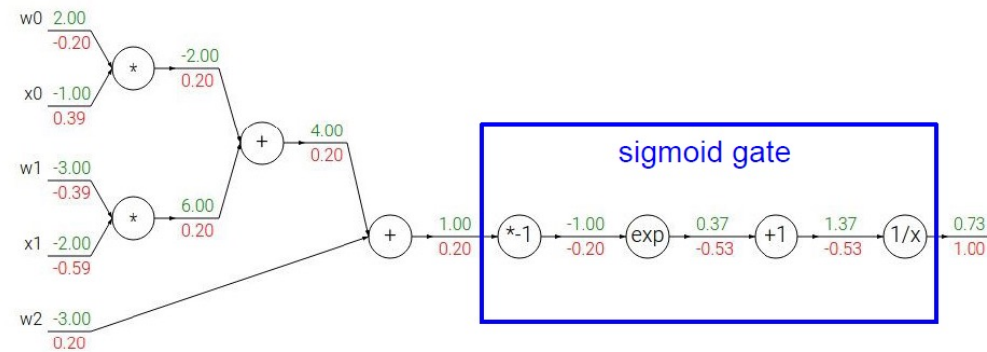
$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Sigmoid

$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad \text{sigmoid function}$$

$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1 + e^{-x})^2} = \left(\frac{1 + e^{-x} - 1}{1 + e^{-x}} \right) \left(\frac{1}{1 + e^{-x}} \right) = (1 - \sigma(x))\sigma(x)$$

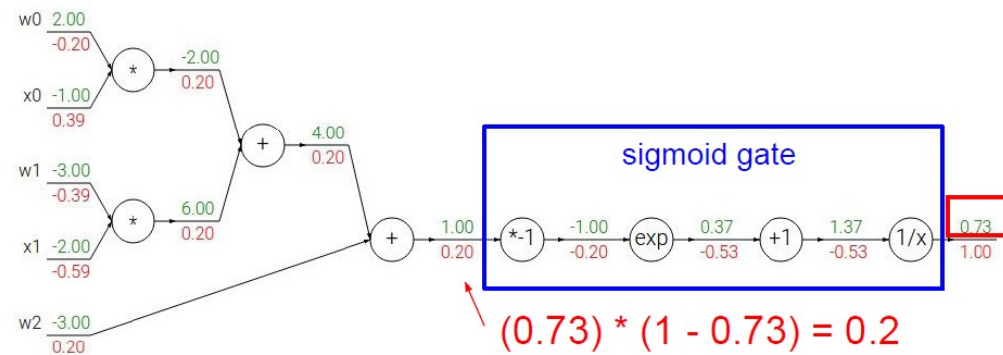


Sigmoid

$$f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad \text{sigmoid function}$$

$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1 + e^{-x})^2} = \left(\frac{1 + e^{-x} - 1}{1 + e^{-x}} \right) \left(\frac{1}{1 + e^{-x}} \right) = (1 - \sigma(x)) \sigma(x)$$



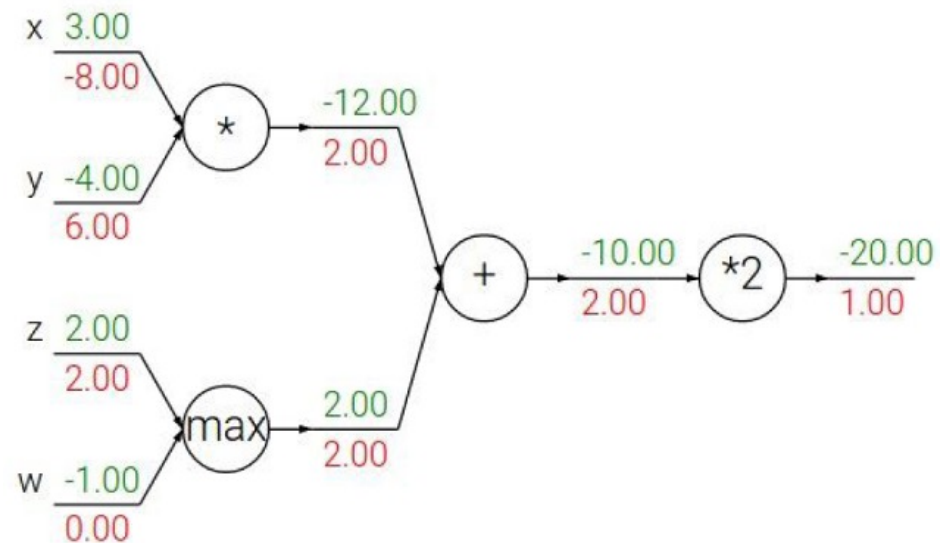
Pattern in backward flow

Patterns in backward flow

add gate: gradient distributor

max gate: gradient router

mul gate: gradient... “switcher”?



Exercise 1

Pooling units take n values x_i , $i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

1. The Lp-pooling module takes positive inputs and computes

$$y = (\sum_i x_i^p)^{\frac{1}{p}}, \text{ assuming we know that } y' = \frac{\partial L}{\partial y}, \text{ what is } x'_i = \frac{\partial L}{\partial x_i} ?$$

2. The log-average module computes $y = \frac{1}{\beta} \ln(\frac{1}{n} \sum_i \exp(\beta x_i))$,

assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x'_i = \frac{\partial L}{\partial x_i}$?

$$q = \frac{1}{n} \cdot \sum_{i=1}^n e^{\beta x_i}$$

$$y = \frac{1}{\beta} \cdot \ln q$$

(1).

$$y = (\sum_{i=1}^n x_i^p)^{\frac{1}{p}}, \quad q = \sum_{i=1}^n x_i^p, \quad y = q^{\frac{1}{p}}$$

$$\textcircled{1} \frac{\partial q}{\partial x_i} = p \cdot x_i^{p-1}$$

$$\textcircled{2} \frac{\partial y}{\partial q} = \frac{1}{p} \cdot q^{\frac{1}{p}-1}$$

$$\textcircled{3} \frac{\partial L}{\partial x_i} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial q} \cdot \frac{\partial q}{\partial x_i} = y' \cdot \frac{1}{p} q^{\frac{1}{p}-1} \cdot p \cdot x_i^{p-1} = y' \cdot (\sum_{i=1}^n x_i^p)^{\frac{1}{p}-1} \cdot x_i^{p-1}$$

$$(2). \quad y = \frac{1}{\beta} \cdot \ln \left(\frac{1}{n} \cdot \sum_{i=1}^n e^{\beta x_i} \right)$$

$$\textcircled{1} \frac{\partial q}{\partial x_i} = \frac{1}{n} \cdot \beta \cdot e^{\beta x_i}$$

$$\textcircled{2} \frac{\partial y}{\partial q} = \frac{1}{\beta} \cdot \frac{1}{q}$$

$$\textcircled{3} \frac{\partial L}{\partial x_i} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial q} \cdot \frac{\partial q}{\partial x_i} \cdot \frac{\partial q}{\partial x_i}$$

$$= y' \cdot \frac{1}{p} \cdot \frac{1}{q} \cdot \frac{1}{n} \cdot \beta \cdot e^{\beta x_i}$$

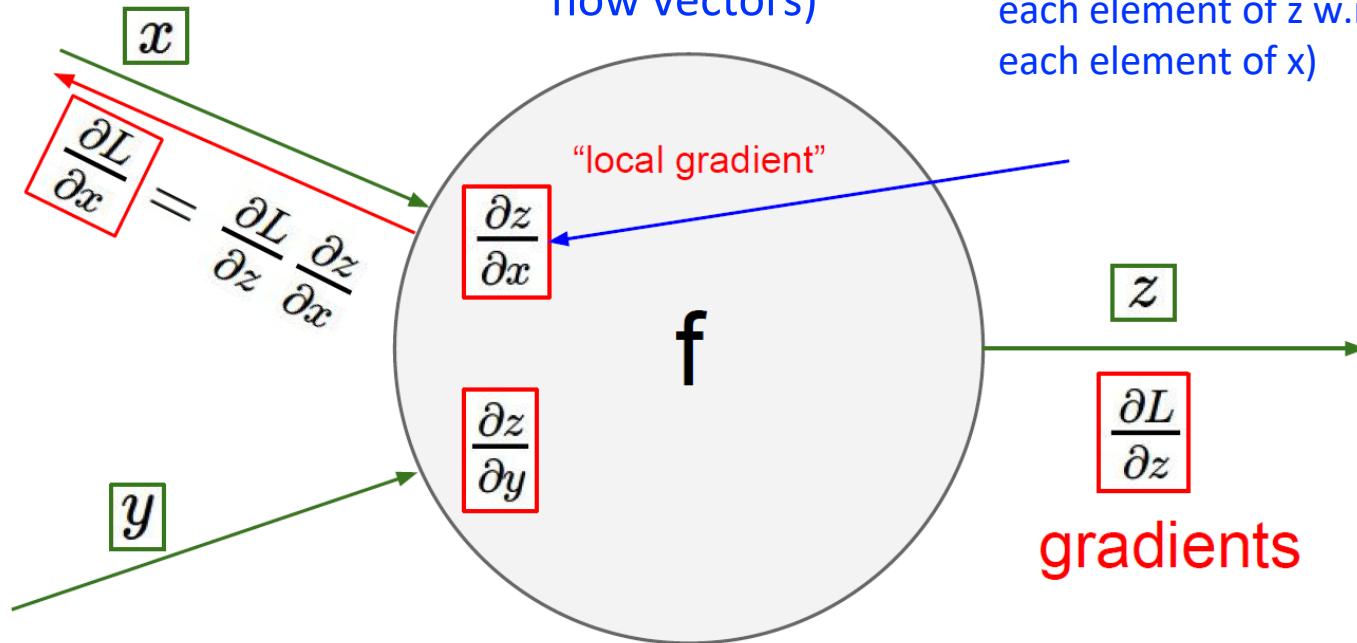
$$= y' \cdot \frac{e^{\beta x_i}}{\sum_{i=1}^n e^{\beta x_i}}$$

Gradients for vector

Gradients for vectorized code

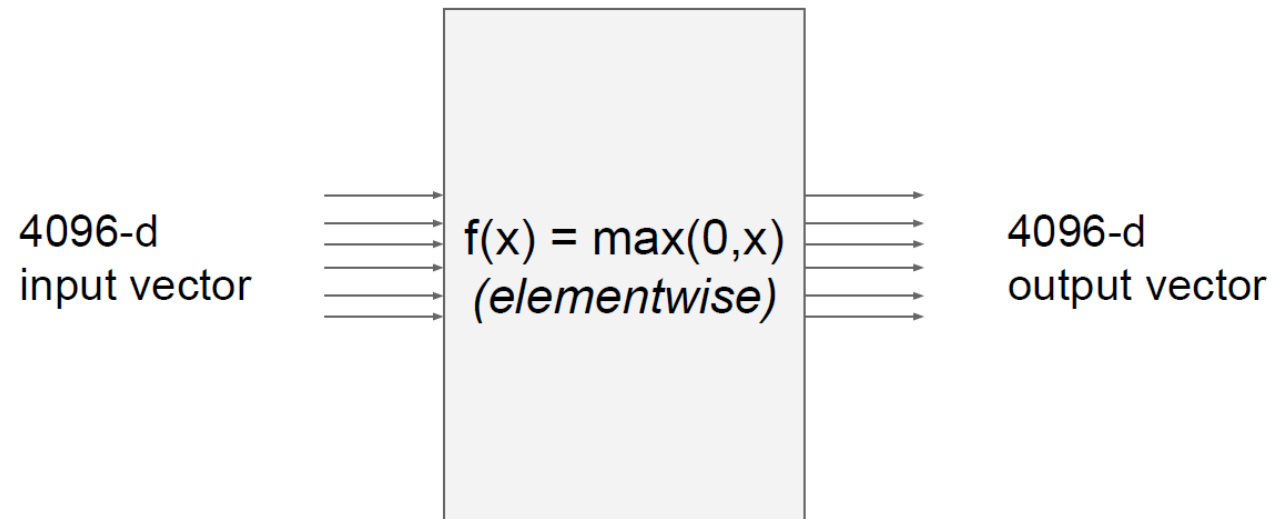
(x, y, z are now vectors)

This is now the **Jacobian matrix** (derivative of each element of z w.r.t. each element of x)



Gradients for vector

Vectorized operation

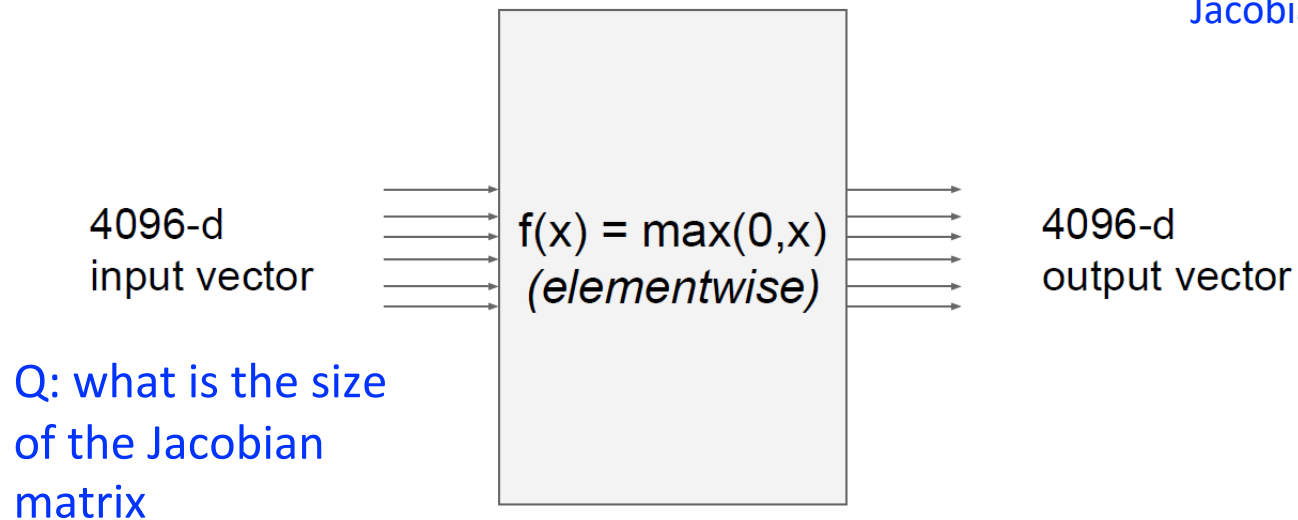


Gradients for vector

Vectorized operation

$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x}} \frac{\partial L}{\partial f}$$

Jacobian matrix



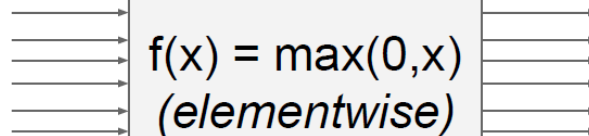
Gradients for vector

Vectorized operation

$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x}} \frac{\partial L}{\partial f}$$

Jacobian matrix

4096-d
input vector



4096-d
output vector

Q: what is the size
of the Jacobian
matrix
[4096 x 4096!]

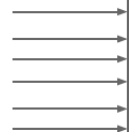
Q2: what does it look
like?

Gradients for vector

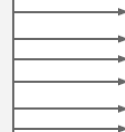
Vectorized operation

in practice we process an entire minibatch (e.g. 100) of examples at one time:

100 4096-d
input vectors



$f(x) = \max(0, x)$
(*elementwise*)



100 4096-d
output vectors

i.e. Jacobian would technically be a
[409,600 x 409,600] matrix :\\

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

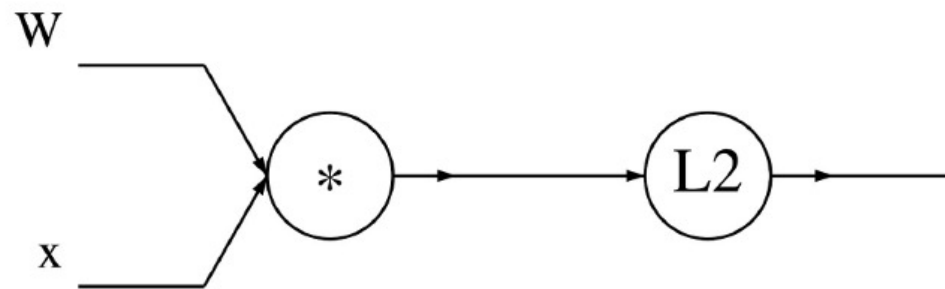
Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{array}{c} \Downarrow \quad \Downarrow \\ \in \mathbb{R}^n \quad \in \mathbb{R}^{n \times n} \end{array}$$

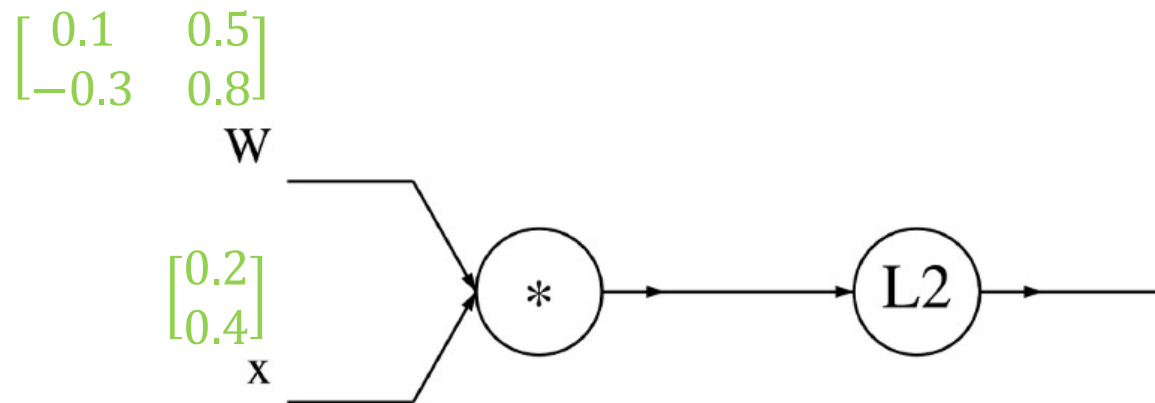
Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



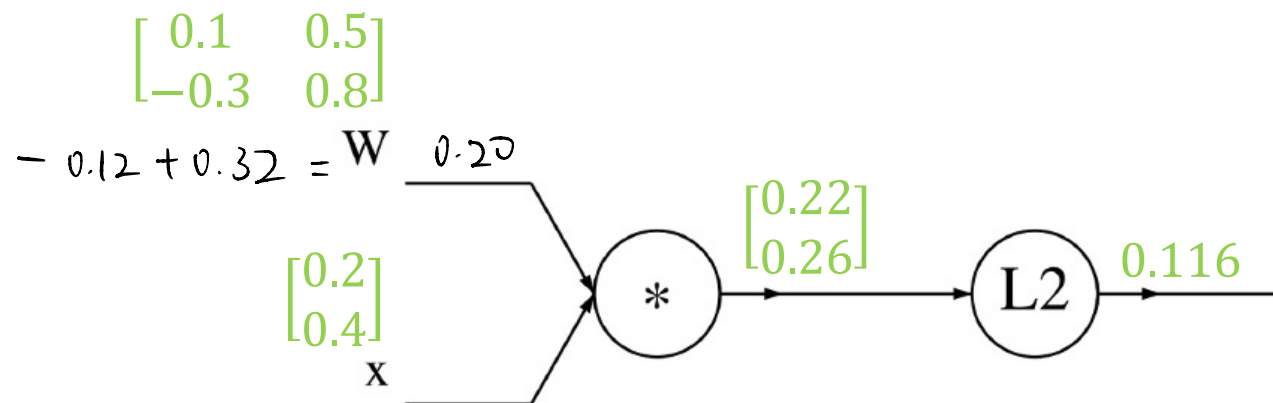
$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \cdots + q_n^2$$

Gradients for vector

$$0.1 \times 0.2 + 0.5 \times 0.4 = 0.02 + 0.2 = 0.22$$

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$f(x, w) = \|W \cdot x\|^2 \quad q = Wx \quad f = q^T q \quad \text{Target: } \frac{\partial L}{\partial q}, \frac{\partial L}{\partial w}, \frac{\partial L}{\partial x}$$

$$(1) \frac{\partial f}{\partial q} = \frac{\partial (q^T q)}{\partial q} = q + (q^T)^T = 2q = \begin{bmatrix} 2q_1 \\ 2q_2 \end{bmatrix} \quad (3) \frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} = W^T \cdot 2q = \begin{bmatrix} W_{11} & W_{21} \\ W_{12} & W_{22} \end{bmatrix} \begin{bmatrix} 2q_1 \\ 2q_2 \end{bmatrix}$$

Gradients for vector

$$(2) \frac{\partial f}{\partial w} = \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial w} = 2q \cdot x^T = \begin{bmatrix} 2q_1 \\ 2q_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} 2q_1 x_1 & 2q_1 x_2 \\ 2q_2 x_1 & 2q_2 x_2 \end{bmatrix} = \begin{bmatrix} 2q_1 W_{11} + 2q_2 W_{21} \\ 2q_1 W_{12} + 2q_2 W_{22} \end{bmatrix}$$

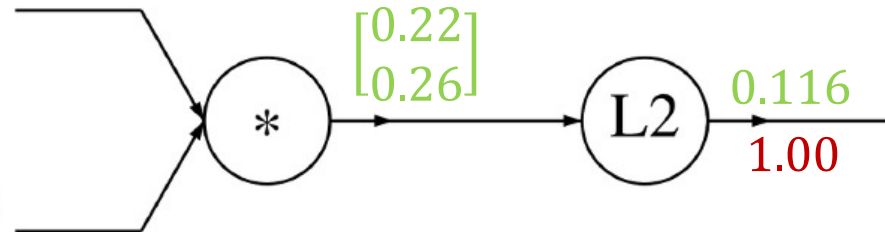
A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

Gradients for vector

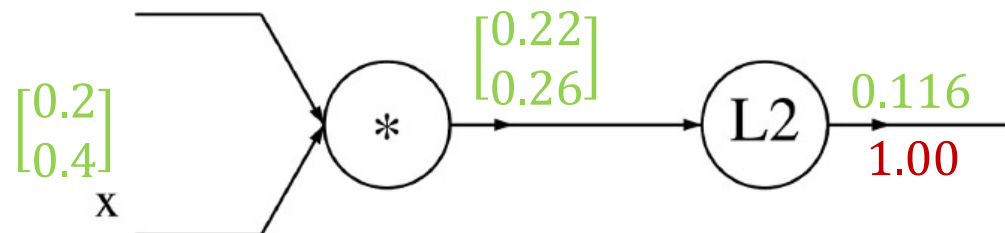
A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \cdots + q_n^2$$

$$\star \left\{ \begin{array}{l} \frac{\partial f}{\partial q_i} = 2q_i \\ \boxed{\nabla_q f = 2q} \end{array} \right.$$

Gradients for vector

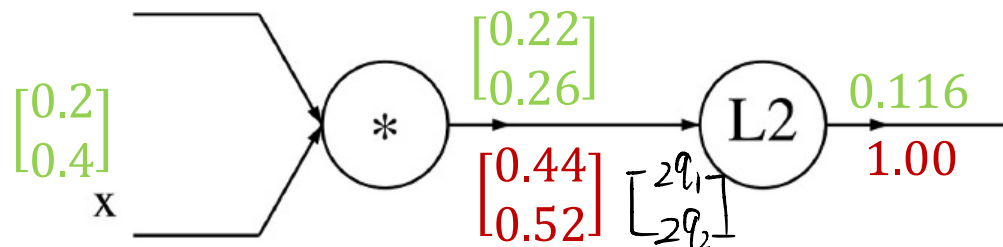
A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\frac{\partial f}{\partial q_i} = 2q_i$$

$$\boxed{\nabla_q f = 2q}$$

Gradients for vector

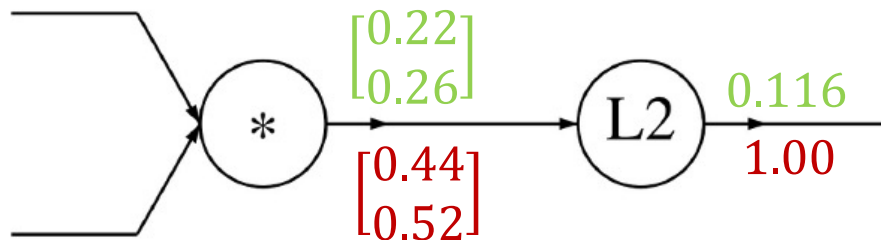
A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

$$f(q) = \|q\|^2 = q_1^2 + \cdots + q_n^2$$

Gradients for vector

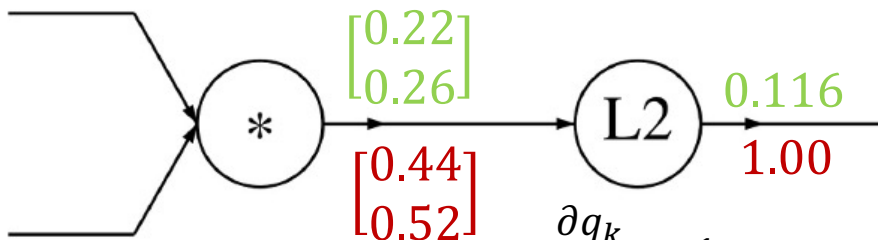
A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

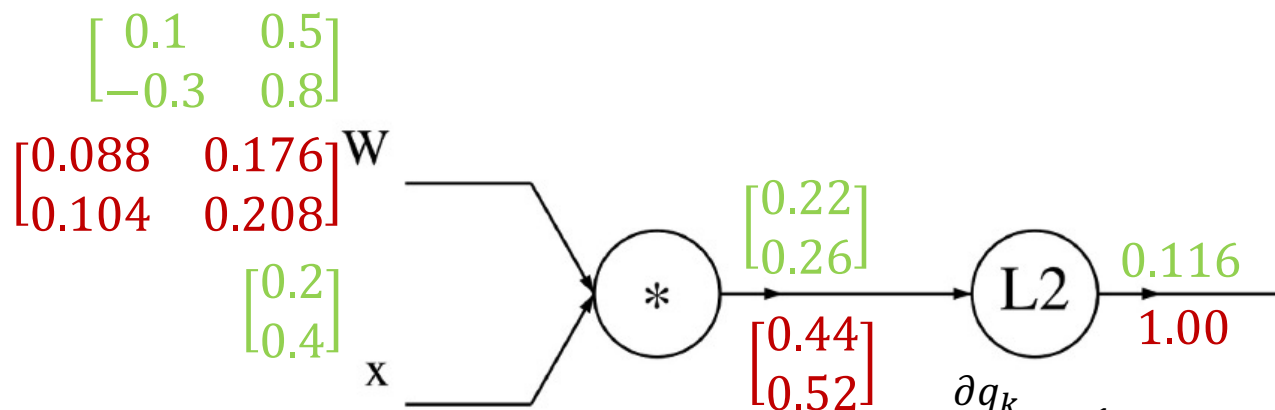
$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i}x_j$$

$$\begin{aligned} \frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\ &= \sum_k (2q_k)(\mathbf{1}_{k=i}x_j) \\ &= 2q_i x_j \end{aligned}$$

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



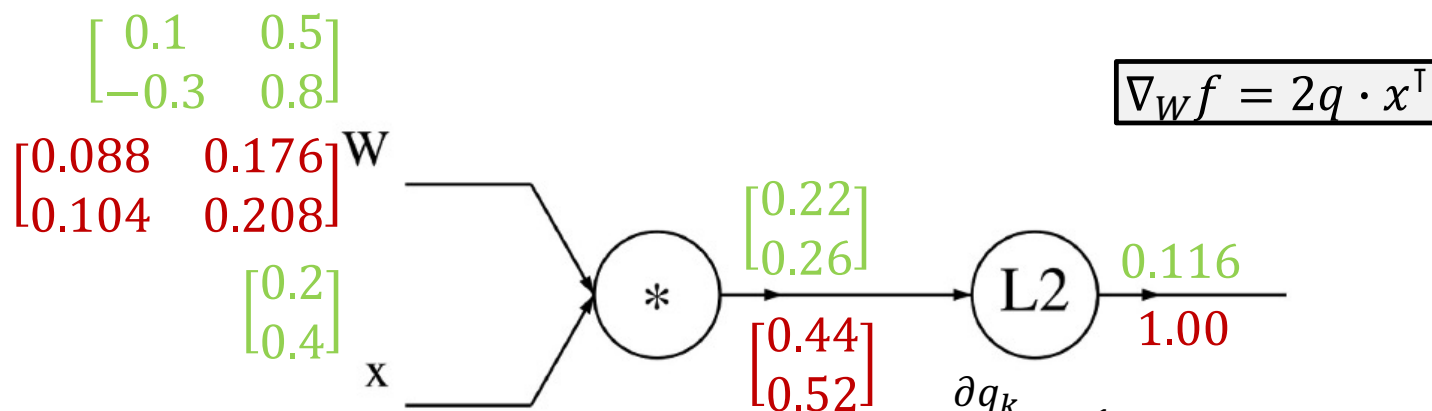
$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\begin{aligned} \frac{\partial q_k}{\partial W_{i,j}} &= \mathbf{1}_{k=i} x_j \\ \frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\ &= \sum_k (2q_k) (\mathbf{1}_{k=i} x_j) \\ &= 2q_i x_j \end{aligned}$$

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



$$\nabla_W f = 2q \cdot x^\top$$

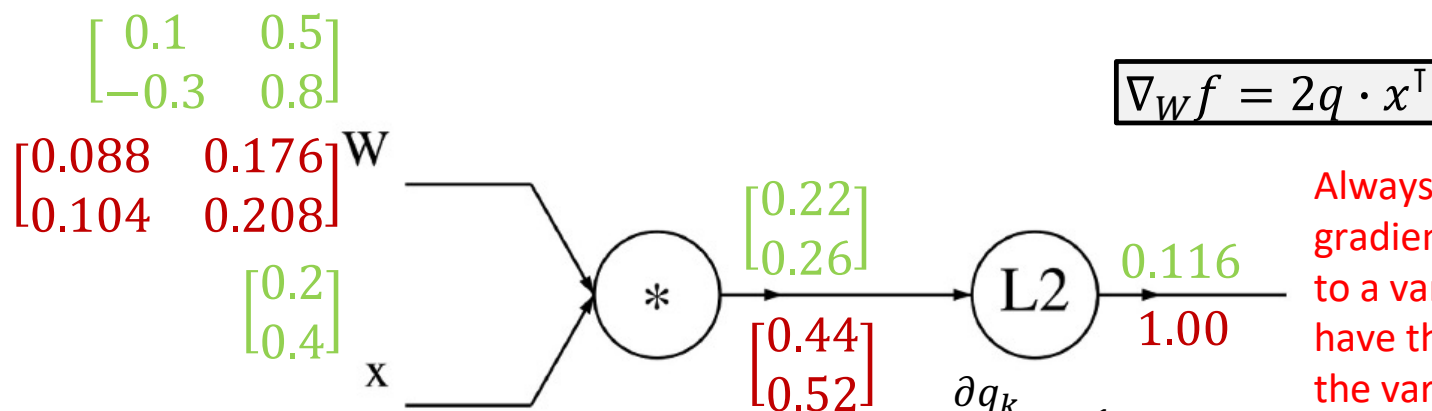
$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\begin{aligned} \frac{\partial q_k}{\partial W_{i,j}} &= \mathbf{1}_{k=i} x_j \\ \frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\ &= \sum_k (2q_k) (\mathbf{1}_{k=i} x_j) \\ &= 2q_i x_j \end{aligned}$$

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



Always check: The gradient with respect to a variable should have the same shape as the variable

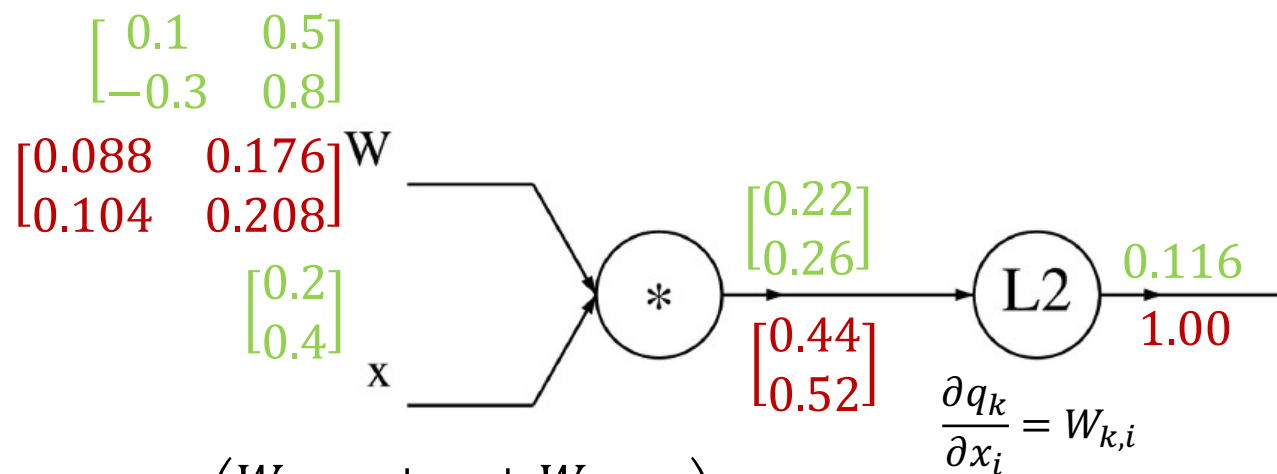
$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\begin{aligned} \frac{\partial q_k}{\partial W_{i,j}} &= \mathbf{1}_{k=i}x_j \\ \frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\ &= \sum_k (2q_k)(\mathbf{1}_{k=i}x_j) \\ &= 2q_i x_j \end{aligned}$$

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

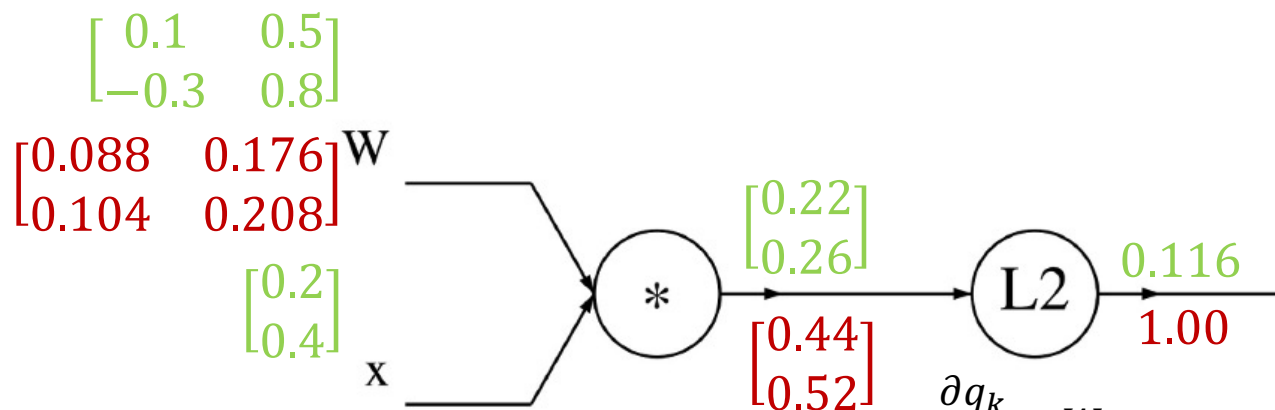


$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \cdots + q_n^2$$

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\frac{\partial q_k}{\partial x_i} = W_{k,i}$$

$$\begin{aligned} \frac{\partial f}{\partial x_i} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial x_i} \\ &= \sum_k 2q_k W_{k,i} \end{aligned}$$

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

